Amplitude-Robust Qubit Spectroscopy Using Echo-Root-Lorentzian Pulse Shapes

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We present a qubit spectroscopy method based on Lorentzian-derived drive pulses that enables T_2-limited frequency characterization at drive amplitudes exceeding those used in conventional spectroscopy by more than three orders of magnitude, thereby enhancing robustness against amplitude variations. Specifically, we employ a root-Lorentzian envelope with a π-phase inversion at its peak, which introduces an echo-like symmetry that cancels on-resonant Rabi oscillations while preserving the spectral response. This approach suppresses conventional power broadening and enables high-resolution spectroscopy using pulses shorter than durations required by standard techniques. We experimentally demonstrate this method on a superconducting qubit platform and observe excellent agreement with numerical simulations, highlighting its efficiency and reliability for rapid and calibration-robust qubit frequency characterization.

INTRODUCTION

Accurate frequency characterization of a quantum two-level system (qubit) is essential for calibrating high-fidelity control in fault-tolerant quantum processors. In many physical implementations, such as superconducting circuits and semiconductor quantum dots, the qubit is realized within an inherently multilevel structure with discrete energy levels, allowing two levels to be selected to form an effective computational subspace that mimics atomic systems via their quantized energy spectrum [1, 2]. Spectroscopic identification of the transition frequency is therefore routinely performed by applying resonant control fields and monitoring the resulting population dynamics [3–5]. Spectroscopic resolution in coherently driven two-level systems is fundamentally constrained by decoherence. In the absence of technical noise, stochastic phase fluctuations induce homogeneous broadening of the transition frequency, leading to a minimum resolvable linewidth that scales inversely with the coherence time T_2, as described by the Bloch equations [6, 7]. This establishes the so-called T_2 limit, which is generally regarded as a fundamental bound on the spectral distinguishability of energy-level transitions.

In practice, however, driven spectroscopy introduces an additional trade-off between signal robustness and spectral resolution. When finite-amplitude control pulses are applied, the system dynamics deviate from the idealized two-level response due to coherent Rabi oscillations and coupling to noncomputational levels. Increasing the drive amplitude improves signal contrast and robustness against state-preparation and measurement errors, but simultaneously induces power broadening, resulting in a linewidth that scales with the instantaneous Rabi frequency. Consequently, conventional spectroscopic techniques must operate at low drive amplitudes in order to maintain resolution approaching the decoherence-limited linewidth set by T_2, rendering precision and robustness fundamentally incompatible.

Ramsey interferometry provides a standard alternative to direct spectroscopy and, in principle, enables frequency estimation at the T_2 limit. Achieving such resolution, however, requires long interrogation times and sufficiently dense temporal sampling to resolve the resulting interference fringes. Moreover, Ramsey measurements are susceptible to systematic biases: asymmetries in the exponential decay envelope—arising, for example, from pulse miscalibration—can shift the apparent resonance peak in the frequency domain. Additional effects such as phase fluctuations or unintended AC Stark shifts during the control pulses introduce further distortions, ultimately biasing the extracted transition frequency.

Continuous efforts aim to mitigate this trade-off through dynamical decoupling or engineered dissipation [8–10]. However, these approaches typically require extended interrogation times or additional control resources, and do not overcome the intrinsic connection between decoherence and spectral resolution, as the same stochastic frequency fluctuations that govern dephasing also determine the resolvability of the transition frequency. Here, we show that this limitation can be bypassed in the driven regime by employing adiabatically engineered Lorentzian-derived drive pulses with a π-phase inversion at their peak. This echo-like structure suppresses on-resonant Rabi oscillations while preserving the steady-state spectral response, enabling T_2-limited frequency characterization at drive amplitudes exceeding those used in conventional spectroscopy by more than three orders of magnitude.

PULSE PROTOCOL AND PHYSICAL MECHANISM

Precise determination of the qubit transition frequency is crucial for implementing high-fidelity quantum gates. The drive frequency must coincide with the qubit resonance, as detuning directly reduces gate fidelity. A rigorous metric for quantifying such calibration errors is the diamond norm, which captures deviations in the imple-